

Manuel Fernández López

Postdoctoral Researcher, Lattice Field Theory Group

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Profile

- Condensed matter theorist specializing in topologically ordered strongly correlated systems and their applications to quantum information science.

Research Focus

- Strongly correlated quantum materials
- Interaction-driven topological transitions
- Green function zeroes as an emergent tool to characterize topological strongly correlated systems

Education

PhD	Autonomous University of Madrid (UAM) , Physics	Madrid, Spain 2019 – 2024
	• Thesis: "Fractionalized topological phases in strongly correlated materials with spin-orbit coupling" • Supervisor: Prof. Jaime Merino • Defense: Oct 2024 — Cum laude	
MSc	Autonomous University of Madrid (UAM) , Condensed Matter Physics and Biological Systems	Madrid, Spain 2017 – 2018
	• Thesis: "Spin-orbit coupling and Coulomb interaction effects in decorated honeycomb lattices"	
BSc	Autonomous University of Madrid (UAM) , Physics	Madrid, Spain 2013 – 2017
	• Theoretical final project: "Bosonic string theory for the derivation of the graviton and compactification of 25+1 dimensions." • Experimental final project: "Data analysis for the determination of the Z₀ boson mass using LHC data." • High honors in Thermodynamic and Statistical Physics I and II, Quantum Mechanics II, Physical Electronics, Magnetism and Superconductivity	

Experience

Institute of Physics, National Yang Ming Chiao Tung University , Postdoctoral Researcher, Lattice Field Theory Group	Hsinchu, Taiwan Sept 2026 – present 1 month
• Member of the Lattice Field Theory Group at NYCU, translating computational techniques between condensed matter physics and non-perturbative QCD. • Applying parton-based and exact diagonalization methods developed for strongly correlated, topologically ordered systems to open questions in hadron structure. • Building complementary expertise in tensor network and DMRG approaches to lattice gauge theories.	

Autonomous University of Madrid (UAM), Associate Researcher

- Developed theoretical approaches to compute Green function zeroes in topological Mott insulators; application to the Kitaev-Hubbard model (preprint in final stage).
- Co-supervised a PhD project on unconventional superconductivity in nickelate heterostructures using semi-analytical and numerical many-body techniques.
- Proposed novel topological strongly correlated systems to apply AI-driven methodologies developed within the group.
- Acted as corresponding author on the first publication of a PhD student, helping to the structuring the manuscript and leading the full reviewer-editor correspondence.

Madrid, Spain
Nov 2024 – Sept 2026
1 year 11 months

Autonomous University of Madrid (UAM), Researcher Formation Fellow (FPI)

- Implemented a finite-T slave-rotor mean-field framework for topological Mott transitions; benchmarked local limits against DMFT.
- Identified flat-band topological superconductivity and localized Majorana edge modes in metal-organic frameworks (MOFs) (Phys. Rev. B 106, 235129, 2022).
- Performed slab single-particle Green function calculations revealing spinon Fermi arcs, a novel type of topological surface state (Phys. Rev. B 109, 085137, 2024).
- Derived effective Hubbard models from first-principles DFT band structures for strongly correlated-material modeling (Phys. Rev. B 111, 165133, 2025).

Madrid, Spain
Sept 2019 – Oct 2024
5 years 2 months

Autonomous University of Madrid (UAM), Scientific Technician, National Research Project

- Adapted master thesis results into a peer-reviewed publication and managed correspondence with editors and referees (Phys. Rev. B 100, 075154, 2019).
- Prepared departmental talks and presented results at a national conference.

Madrid, Spain
Mar 2018 – Aug 2019
1 year 6 months

***Research Project* • Institute of Physics, NYCU (NSTC grant 112-2112-M-A49-021-MY3), PI: Prof. C.-J. David Lin, Tensor networks and quantum computing for high-energy physics**

- Contributing to a project applying tensor-network and quantum-computing methods, developed for strongly correlated condensed matter systems, to problems in high-energy physics.

Lattice Field Theory Group,
NYCU
2026 – present
1 year

***Research Project* • Dept. of Theoretical Condensed Matter Physics, UAM (Ref. PID2022-139995NB-I00), New concepts and methods for topological strongly correlated quantum matter**

- Computed non-local Green's function in-gap zeros delineating a fractionalized protected state (spinon drumhead states).
- Led a research line computing Green's function zeros in Kitaev materials such as α - RuCl_3 , opening a route to identify topological phases in correlated superconductors.

IP: Jorge Bravo Abad
2023 – 2026
3 years

***Research Project* • Dept. of Theoretical Condensed Matter Physics, UAM (Ref. RTI2018-098452-B-I00), New phenomena and applications of topological strongly correlated materials in and out of equilibrium**

- Supervised a master's student on code adaptation for correlated nodal-line semimetals, providing the first theoretical realization of the observed pressure-induced Mott transition.
- Developed a semi-analytical technique extended to finite temperature revealing that non-local correlations lead to spinon Fermi arcs, with predicted ARPES signatures on pyrochlore iridates.
- Investigated correlated topology in flat-band decorated honeycomb lattices motivated by MOFs; exact diagonalization revealed a flat-band quantum spin liquid consistent with experiments.
- Explored flat-band unconventional superconductivity from doping this spin liquid, characterizing localized Majorana edge modes and their potential for topological quantum computing.

IP: Jaime Merino Troncoso
(FPI PRE2019-087603)
2019 – 2021
2 years

Research Project • Dept. of Theoretical Condensed Matter Physics, UAM (Ref. MAT2015-66128-R) , New electronic and photonic materials based on topological strongly correlated systems	IP: Jaime Merino Troncoso 2016 – 2019 3 years
- Implemented Hartree-Fock calculations in a decorated honeycomb lattice relevant to certain MOFs, identifying topological semimetals. - Presented findings at national conferences and acted as corresponding author in the peer-review process.	
Teaching • Autonomous University of Madrid (UAM) , Co-supervisor of a Master's and current Ph.D. student	Madrid, Spain 2024 – present 2 years
- Mentoring on topological strongly correlated electron models and Green function zeroes analysis, resulting in a joint publication. - As corresponding author on the student first paper, guided the full publication pipeline (submission, revision, peer-review dialogue) while enforcing good practices in code/documentation and reproducible workflows.	
Teaching • Autonomous University of Madrid (UAM) , Lecturer & Lab Supervisor, Physics Laboratory course	Madrid, Spain 2021 – 2022 1 year
Physics Laboratory course for Environmental Science undergraduates (2nd year).	

Peer-Review Activity

- Referee report prepared for Physical Review Letters (submitted Aug 2025; editorial decision Sep 2025), assessing significance, methodological soundness and clarity of presentation.
- Regularly manage reviewer/editor correspondence for co-authored and first-authored manuscripts, including coordinating responses, revising manuscripts, and ensuring reporting/figures meet journal standards.
- Peer reviewer for Physical Review B, first as co-referee with supervisor, later as an independent referee.
- Experience assessing reproducibility, figure clarity, methodological soundness and experimental relevance in theoretical manuscripts, with a focus on transparent availability of data and custom code.

Presentations

- From Quantum Anomalous Hall Phases to Topological Metals in Interacting Decorated Honeycomb Lattices* — Oral contribution, INC Young Researchers Meeting, Madrid (2019)
- Magnetism and Topological Phases in an Interacting Decorated Honeycomb Lattice with Spin-Orbit Coupling* — Oral contribution, INC Young Researchers Meeting, Madrid (2021)
- Topological Phases Induced by Interactions and Spin-Orbit Coupling in Decorated Honeycomb Lattices* — Oral contribution, APS March Meeting, online (2021)
- Bad Topological Semimetals in Layered Honeycomb Compounds* — Oral contribution, APS March Meeting, Chicago, IL, USA (2022)
- Bad Topological Semimetals in Layered Honeycomb Compounds* — Invited talk, School of Mathematics and Physics, University of Queensland, Australia (2022)
- Emergence of Spinon Fermi Arcs in the Weyl-Mott Metal-Insulator Transition* — Oral contribution, INC Young Researchers Meeting, Madrid (2024)
- Bad Weyl Semimetals and Spinon Fermi Arcs in a Model for Pyrochlore Iridates* — Oral contribution, IOP CMQM 2024, St. Andrews, Scotland (2024)

Technical Skills

- **Semianalytical / Numerical:** Parton mean-field theories, exact diagonalization, solid working knowledge of (cluster) DMFT and QMC methods, topological invariants calculation in the absence of Landau quasiparticles
- **Programming & Software:** LaTeX, Fortran, Matlab (high proficiency); Adobe Suite, Wolfram Mathematica (advanced); Python (beginner); Windows, macOS, Linux
- **Communication:** Manuscript evaluation, referee correspondence, figure/table preparation
- **Languages:** English (fluent), Spanish (native)

Honors & Fellowships

- **FPI Research Fellowship**, Autonomous University of Madrid (UAM) (2019–2024)